

ZAREWA INDUSTRIES

Roofing & Building Materials Manufacturing

SYSTEM OF OPERATIONS

Complete Procedures, Policies & Standards — Code-Accurate Edition v2.0

Generated 24 June 2026

Branch	Code	Role	Status
Kaduna (HQ)	KD / BR-KD	Headquarters & Sales	Active — System Default
Yola	YL / BR-YOL	Manufacturing Factory	Active
Maiduguri	MDG / BR-MAI	Manufacturing Factory	Active

1. COMPANY OVERVIEW & IDENTITY

Zarewa Industries manufactures and sells roofing sheets, stone-coated tiles, and building accessories. Three active sites are managed through a single integrated ERP system.

1.1 Products

- Long-span & step-tile aluminium/aluzinc roofing sheets (multiple gauges & profiles)
- Stone-coated steel roofing tiles (stone-flatsheet format)
- Roofing accessories — ridges, flashings, gutters, fixings
- Outside corrugation (sub-contracted to third parties)
- Transport & installation services (quoted separately on quotation)

1.2 Document ID Format: PREFIX-BRANCHCODE-YY-NNNN

All documents carry a human-readable ID. Sequences reset each calendar year per branch. Legacy IDs without branch code are still recognised to avoid collisions.

Prefix	Document Type	Example
QT	Sales Quotation	QT-KD-26-0001
LE	Ledger Entry / Receipt	LE-KD-26-0001
RCP	Confirmed Sales Receipt	RCP-KD-26-0001
DLV	Delivery Note	DLV-KD-26-0001
CL	Cutting List (Production Order)	CL-KD-26-0001
PJ	Production Job	PJ-KD-26-0001
PO	Purchase Order	PO-KD-26-0001
EXP	Expense Record	EXP-KD-26-0001
TM	Treasury Movement	TM-KD-26-0001
SM	Stock Movement	SM-KD-26-0001
BD	Bank Deposit	BD-KD-26-0001
BDA	Bank Deposit Allocation	BDA-KD-26-0001
WI	Office Work Item	WI-26-0001
MACH	Machine Asset	MACH-26-0001
MXPL	Maintenance Plan	MXPL-26-0001
MXWO	Maintenance Work Order	MXWO-26-0001
MXEV	Maintenance Event	MXEV-26-0001
CREV	Coil Control Event	CREV-KD-26-0001
CREQ	Coil Request	CREQ-KD-26-0001
IT	In-Transit Load	IT-KD-26-0001
MINT	Material Incident	MINT-KD-26-0001
MREQ	Material Request	MREQ-KD-26-0001
OBL	Staff Obligation Account	OBL-KD-26-0001

2. ERP SYSTEM ARCHITECTURE & SECURITY

2.1 Technology Stack

Layer	Technology	Notes
Backend	Node.js + Express 5	REST API, server-side sessions, CSRF enforcement
Frontend	React (Vite SPA)	Role-gated module routing via ModuleRouteGuard & FinanceDeskRouteGuard
Database	SQLite via better-sqlite3	124+ tables; synchronous write transactions for integrity
Authentication	Session cookies + CSRF tokens	bcrypt password hashing; sliding session timeout

2.2 Authentication & Session Controls

Control	Value / Rule
Password minimum length	8 characters
Password complexity	Uppercase letter + lowercase letter + digit + special character (all required)
Session timeout	120 minutes sliding (env SESSION_TIMEOUT_MINUTES, range 5-480)
Failed login lock threshold	5 consecutive failures (env FAILED_LOGIN_LOCK_THRESHOLD, default 5)
Lock duration	30 minutes (env ACCOUNT_LOCK_MINUTES, default 30)
CSRF protection	Token validated on every mutating request
Must-change password	Supported — user forced to change on next login after admin reset

[CRITICAL] After 5 failed logins the account is locked for 30 minutes. Only Admin or MD can unlock accounts. Branch Managers cannot reset passwords.

2.3 Branch Data Isolation

Every API request is scoped to the authenticated user's assigned branch. Cross-branch rollup is available only to MD, CEO, Chairman, and Admin. Finance and Sales staff see only their own branch data unless granted explicit override permission.

2.4 Period Control

All financial writes (receipts, expenses, GL journals, treasury movements) enforce `assertPeriodOpen()`. Posting to a closed accounting period is rejected with a hard error. Period management is the responsibility of the `finance_manager` role.

2.5 Edit Approval System

Changes to confirmed/settled documents require an Edit Approval request — logged, sent to Branch Manager or MD, and applied only on approval. Fully audited in the `approval_actions` table. Rejected edits are recorded with the reason.

2.6 Database Table Groups (124+ tables)

Domain	Key Tables
Sales & Quotations	quotations, cutting_lists, deliveries, sales_receipts, customer_refunds, customer_crm_interactions
Ledger & Finance	ledger_entries, expenses, payment_requests, treasury_accounts, treasury_movements
GL / Accounting	gl_journal_entries, gl_journal_lines, gl_accounts, accounting_register_settlements
Procurement	purchase_orders, procurement_catalog, suppliers, coil_lots, yard_coils, coil_requests
Production	production_jobs, wip_balances, in_transit_loads, production_completion_adjustments, production_job_stone_flatsheet_usage
Inventory	stock_movements, product_branch_inventory, material_requests, material_incidents, incident_registry
Bank & Treasury	bank_deposits, bank_deposit_allocations, bank_reconciliation_lines, inter_branch_loans
Fixed Assets & Machines	fixed_assets, machines, machine_asset_links, machine_meter_logs, maintenance_plans, maintenance_work_orders, maintenance_events, maintenance_cost_lines
HR Core	hr_staff, hr_request_leave, hr_request_loan, hr_leave_balances, hr_leave_accrual_ledger, hr_request_discipline
HR Payroll & Benefits	hr_payroll_line_loans, hr_staff_obligation_accounts, hr_staff_obligation_transactions, hr_employment_letters, hr_chairman_school_fees, hr_chairman_expenses
HR Governance	hr_performance_reviews, hr_policy_acknowledgements, hr_audit_events, hr_id_cards, hr_attendance_uploads, hr_daily_roll_calls
Office & Communication	work_items, work_item_decisions, official_notices, forum_topics, forum_posts, office_dossiers, work_item_filing, work_item_print_snapshots
Credit & Staff Obligations	credit_exceptions, hr_staff_obligation_accounts, hr_staff_obligation_transactions
AI / Help (Zare)	help_rag_chunks, help_user_memory, help_branch_memory, help_workflow_events, help_knowledge_gaps, help_article_weights
System	org_policy_kv, org_policy_audit, human_id_sequences, audit_log, approval_actions, reference_counters

3. USER ROLES, PERMISSIONS & ACCESS CONTROL

[IMPORTANT] The role key "finance_manager" displays in the UI and documentation as "Accountant / Head of Accounts" — confirmed in auth.js. The label "Finance Manager" in older documentation is incorrect.

3.1 Role Catalogue

Role Key	Display Title	Function
admin	System Administrator	Full system access. User management, settings, data resets. Cannot be locked out.
md	Managing Director	All operational approvals. Treasury reserve policy. Notices management. Accounting & cashier desk access by default. All-branch rollup.
ceo	Chief Executive Officer	Same permission scope as MD. Executive overview, sensitive payroll view, staff directory.
chairman	Chairman (Board)	Identical to CEO permissions. Additionally manages family/domestic accounts and scholarship payments. Present in auth.js — absent from legacy documentation.
finance_manager	Accountant / Head of Accounts	Accounting desk, GL journals, reconciliation, trial balance, payables, fixed assets, AP phases, period management.
cashier	Cashier	Cashier desk only (Phase B separation). Receives payments, confirms receipts. Cannot access Accounting desk.
sales_manager	Branch Manager	Branch management: quotation approval, cutting list release, expense approval, team HR, delivery management.
sales_staff	Sales Staff	Create quotations, receive payments (if permitted), manage customer interactions, CRM.
operations_officer	Operations Officer / Storekeeper	Inventory, deliveries, production dispatch, stock register. Normalised from legacy roles storekeeper and store_keeper.
hr_admin	HR Administrator	Full HR: onboarding, payroll processing, leave management, discipline cases, exit clearance, asset custody tracking.
gmhr	GM, Human Resources	HR final approvals: leave, loans, transfers, exit final clearance, employment letter signing.
hr_portal_only	HR Portal (Self-Service Only)	Mining division / scholarship staff. Self-service ONLY — cannot access any operational ERP module.

3.2 Finance Desk Separation (Phase B)

The Cashier and Accounting desks are physically separated by permission and the FinanceDeskRouteGuard in the frontend. This is a core internal control preventing one person from both receiving payments and reconciling accounts.

Desk	URL	Permission	Capabilities
Cashier Desk	/cashier	cashier.desk.view + cashier.receipts.confirm	Receive payments, confirm receipts, manage cashier queue, process transfers
Accounting Desk	/accounting	accounting.desk.view	GL journals, trial balance, reconciliation, bank deposits, fixed assets, debtors/creditors, period close

3.3 Key Permission Groups

Group	Key Permissions	Default Roles
Sales	sales.view, sales.approve, quotation.create, quotation.approve	sales_manager, sales_staff, md, admin
Finance — Cashier	cashier.desk.view, cashier.receipts.confirm	cashier, md
Finance — Accounting	accounting.desk.view, accounting.reconciliation.view, accounting.gl.view	finance_manager, md, admin
Finance Actions	finance.post, finance.approve, finance.pay	finance_manager, admin
Treasury	treasury.reserve_policy.manage	md, admin
Office	office.use (all staff), notices.manage (senior only)	All roles; notices.manage = md/admin/hr_admin/gmhr/ceo/chairman
Reports	reports.view, reports.export	finance_manager, sales_manager, md, admin
HR — Self-Service	hr.self, my_profile.view, my_leave.request, my_loan.request, my_attendance.view, my_payslip.view	All roles including hr_portal_only
HR — Team	hr.team.view, attendance.mark, daily_roll.mark, leave.endorse, loan.endorse	sales_manager, hr_admin
HR — Admin	hr.staff.manage, all 5 exit stages, assets.custody.manage, hr.recovery.manage	hr_admin
HR — GMHR	gm_approve, final_approve, exit.final_clear, letters.approve	gmhr
HR — Executive	executive.view, hr.payroll.md_approve, salary_structure.approve, hr.chairman.manage	md, ceo, chairman
HR — Finance	hr.payroll.pay, hr.payroll.export, hr.loan.disburse, hr.exit.finance_clear	finance_manager (financeHr bundle)

3.4 Module Visibility by Role

URL / Module	Who Can Access
/sales	All roles except hr_portal_only
/customers	sales_manager, sales_staff, finance_manager, md, admin
/procurement	operations_officer, sales_manager, finance_manager, md, admin
/operations	operations_officer, sales_manager, md, admin
/cashier	cashier (cashier.desk.view), md
/accounting	finance_manager (accounting.desk.view), md, admin
/accounts / bank-reconciliation	finance_manager, md, admin
/reports	finance_manager, sales_manager, md, admin
/analytics	finance_manager, sales_manager, md, admin, ceo, chairman
/exec	md, ceo, chairman, admin
/office	All staff with office.use
/hr/*	hr_admin, gmhr
/executive-hr/*	md, ceo, chairman
/team-hr/*	sales_manager
/me/* or /my-profile/*	All staff (self-service portal)
/price-list	md, admin, sales_manager
/pricing-policy	md, admin
/edit-approvals	sales_manager, md, admin
/settings	admin, md (limited)

4. SALES & CUSTOMER MANAGEMENT

4.1 Customer Records

Customers are registered as master records. Phone numbers are normalised to a canonical key (customerPhoneKey.js) and emails are similarly normalised for duplicate detection. Customer CRM interactions (calls, site visits, follow-ups) are logged against the customer and optionally against a specific quotation.

4.2 Pricing

1. Standard price list (maintained by MD/Admin at /price-list) — base price per metre/sheet per gauge and profile.
2. Pricing policy rules (/pricing-policy) — tiered pricing, minimum floor prices, exception approval routing.
3. Quotation-level overrides — sales staff can enter a custom price, but prices below the approved floor trigger a "Below Floor" exception requiring Branch Manager or MD approval before quotation activates.

[NOTE] Price snapshots are captured at quotation creation. Subsequent price list changes do not retroactively affect confirmed quotations.

4.3 Credit Exception System

Default policy: all sales require advance or cash payment. Customer credit requires a Credit Exception:

Approval Level	Condition	Configuration
Branch Manager	Amount below BM limit	CREDIT_BRANCH_MANAGER_LIMIT_NGN (env or org_policy_kv)
MD	Amount above BM limit	CREDIT_MD_REQUIRED_ABOVE_NGN (env or org_policy_kv)
MD (always)	No credit policy configured	Default fallback until policy is configured in org settings

Field	Details
Status flow	pending → approved / rejected / revoked / expired
Default credit terms	14 days (configurable)
Maximum credit terms	90 days (configurable)
Effect on delivery gate	Active approved exception bypasses the payment enforcement gate for delivery

4.4 Delivery Payment Gate

Mode	Behaviour	Env Setting
off	No payment check — delivery always allowed	DELIVERY_PAYMENT_GATE=off
warn	Warning shown to staff but delivery still proceeds	DELIVERY_PAYMENT_GATE=warn
enforce	Delivery BLOCKED until balance cleared or active credit exception exists	DELIVERY_PAYMENT_GATE=enforce

[IMPORTANT] ALLOW_MD_DELIVERY_OVERRIDE=true lets MD bypass enforce mode per delivery. DELIVERY_PAYMENT_GATE_STRICT_FINANCE additionally requires Finance team confirmation before dispatch.

4.5 Customer Refunds — Reason Categories

Category	Typical Cause
Order cancellation	Customer cancels before production starts
Unproduced meterage	Production not completed for ordered quantity
Overpayment	Customer paid more than invoice value
Transport issue	Delivery problem attributed to logistics
Installation issue	Defect discovered during or after installation
Additional services	Credit for agreed extra work provided to customer
Accessory shortfall	Accessories not included with delivery
Stone flatsheet shortfall	Stone tiles short on delivery
Calculation error	Pricing or quantity mistake on original quotation
Substitution Difference	Gauge or profile was substituted; price difference refunded
Customer commission	Sales agent commission credited back to customer
Other	Catch-all; requires written justification

[NOTE] Minimum refundable balance to trigger refund workflow: ₦1,000. Refund payer and approver must be different individuals (assertRefundPayerNotApprover).

5. QUOTATION & ORDER LIFECYCLE

5.1 Quotation Status Flow

Status	Meaning	Triggered By
draft	Created, not yet submitted	Sales staff
pending_approval	Submitted — below-floor price exception awaiting approval	Sales staff
active	Confirmed — production can be scheduled	Branch Manager (or MD if price exception)
cutting_list_issued	Cutting List generated and sent to factory	Branch Manager / Operations
in_production	Factory currently producing	Production (cutting list status update)
completed	All production and delivery confirmed	System — on full delivery confirmation
cancelled	Voided — refund workflow triggered if payments received	Branch Manager / MD

5.2 Production Payment Gate (Cutting List Release)

Before a Cutting List can be issued to the factory, the quotation must have received a minimum payment fraction. Default threshold: 70% of invoice value (configurable per branch via `cutting_list_min_paid_fraction` in the branches table).

[NOTE] Branch Manager can grant a production gate override with documented justification. The override is logged and visible to the MD in the Executive Dashboard decision queue.

5.3 Staff Purchase Credit

Staff members may purchase roofing materials on personal credit — a separate workflow from customer sales:

Policy Rule	Value
Minimum service for eligibility	1 year of employment
Maximum outstanding obligation	₦5,000,000
Maximum per-purchase amount	₦2,000,000
Maximum repayment period	12 months
Maximum concurrent active credits	1 at a time
Obligation account ID format	OBL-{BranchCode}-{YY}-{seq}
Repayment method	Monthly payroll deduction (<code>hr_payroll_line_loans</code>) + optional direct cash repayment at cashier

5.4 Accessory & Stone Flatsheet Fulfillment

Accessories (ridges, gutters, fixings) and stone flatsheet items are tracked and fulfilled separately from roofing sheet production. Stone items are converted from sheet count to m² (`stoneFlatsheetSheetsToM2`). Partial fulfillment is supported across multiple deliveries.

5.5 In-Transit Inventory

Production-completed goods moving from factory to branch are tracked as In-Transit Loads (IT-XX-YY-NNNN). They appear in inventory reports and are reconciled during the monthly stock register close process.

6. PRODUCTION OPERATIONS

6.1 Production Workflow

Step	Action	System Event
1	Sales issues Cutting List to factory	CL status → issued; Production Job (PJ) created
2	Coil assigned to job from coil_lots inventory	Coil reserved; coil_control_event (CREV) logged
3	Production Manager starts job	PJ status → in_production; WIP balance opened
4	Rolling/corrugation completed; metres measured	Production metres recorded; conversion variance computed
5	Production Manager completes job	PJ → completed; Revenue recognised (AP1c); GL auto-posted
6	Quality Control review (if required)	QC flag attached to PJ; QCO sign-off recorded
7	Operations packs and dispatches delivery	Delivery (DLV) created; stock deducted; delivery gate checked
8	Customer signs delivery note	DLV confirmed; Branch Manager or Operations confirms receipt

6.2 Revenue Recognition — Policy V1 / AP1c

Under Policy V1 (the active accounting policy), revenue is recognised at production completion — NOT at delivery or invoice date. The AP1c module posts GL entries automatically on job completion:

- Dr: Accounts Receivable (customer ledger)
- Cr: Sales Revenue (GL account 4000)

[IMPORTANT] Revenue is on the books the moment the factory completes production, even before physical delivery. This is the most significant policy difference from standard invoicing-basis accounting.

6.3 Coil Management

Attribute	Description
Coil number	Manufacturer serial — unique identifier for each roll of steel
Material spec	Gauge (mm thickness), colour, design/profile, type (aluzinc or aluminium)
Weight tracking	Gross weight at receipt; remaining weight tracked through each production job
production_blocked flag	Blocks job assignment; block reason required; MD auto-notified on coil short-receipt
Coil control events (CREV)	Every coil movement, inspection, and control action logged immutably
Tail finish threshold	85 kg — when remaining weight ≤ 85 kg, system flags "Roll Finished" and clears residual stock

6.4 Production Conversion Variance

When actual output metres differ from theoretical (based on standard kg/m conversion per gauge), the Production Manager must select a documented variance reason before completing the job. Accepted reasons include: material defect, coil splicing, cutting error, specification change, calibration difference.

6.5 WIP Balances

Work-in-Progress balances track cost of goods in production. Job start: coil cost debited to WIP. Job completion: WIP cleared and transferred to COGS under the AP3 material cost classification framework.

6.6 Material Incidents

Type	Description
missing_asset	Tool or equipment reported missing
coil_damage	Coil damaged on receipt or in storage
production_offcut	Significant cutting waste beyond normal tolerance
asset_loss / missing_pump	Equipment loss (aliases mapped to missing_asset)

Incidents are registered in incident_registry, linked to discipline cases if staff negligence is established, and recovery schedules can be generated to deduct costs from salary via payroll.

7. PROCUREMENT & INVENTORY

7.1 Purchase Order Types (derived from line items in poLineTypes.js)

PO Kind	Description
Coil procurement	Steel coil purchase; coil_lots record created on GRN
Accessories	Consumables, fixings, ridges, gutters for resale/production
Stone/flatsheet material	Stone aggregate or flatsheet substrates for stone-coated tile production
Supplier advance	Pre-payment to supplier before goods receipt (tracked under AP2)
Mixed PO	Combination of the above types in a single order

7.2 Goods Receipt (GRN) Process

1. Coil arrival recorded in coil_lots with weight, gauge, colour, and coil number.
2. Inventory GL journal posted: Dr Inventory (1300), Cr Accounts Payable (2000).
3. Coil linked to purchase order; landed cost (including carriage inward) recorded.
4. If short-receipt detected (quantity short vs. PO), MD is automatically notified (notifyMdCoilShortReceipt).

[NOTE] Under AP2 (Received-Basis), the GRN triggers the payables liability. Under the prior policy, payables were recognised at payment date.

7.3 Monthly Stock Register — 4-Stage Sign-Off

Stage	Action	Owner	System Status After
1 — Store Confirmation	Physical stock counts matched to system records	Operations Officer	store_confirmed
2 — Branch Manager Approval	BM reviews and approves; BM adjustment window opens	Branch Manager	bm_approved
3 — MD Approval	MD reviews all branches' stock registers	Managing Director	md_approved
4 — Lock	Register locked; print snapshot captured; no further edits possible	Admin / MD	locked

7.4 Suppliers & Transport Agents

Suppliers and transport agents are global master data — shared across all branches. Supplier identity is deduplicated using tax ID and phone normalisation. Transport agents are tracked separately and can be linked to individual deliveries.

7.5 Inter-Branch Material Requests

Branches request materials from each other via Material Requests (MREQ-XX-YY-NNNN). Fulfilled requests create stock movement records and, where applicable, inter-branch loan entries in the treasury.

8. FINANCE — CASHIER DESK

8.1 Payment Receiving Process

1. Customer pays by cash, bank transfer, or cheque.
2. Sales staff records the receipt against the quotation reference.
3. Cashier confirms the receipt (cashier.receipts.confirm permission required).
4. Confirmed receipt enters the ledger; GL auto-posted (AP1c).
5. Treasury movement recorded in the appropriate account (cash safe or bank account).

[CRITICAL] Only confirmed receipts are recognised in the ledger and financial reports. Unconfirmed receipts remain in the cashier pending queue invisible to reports.

8.2 Receipt Types & GL Treatment

Type	Description	GL Entry
Customer Receipt	Payment against outstanding balance on a quotation	Dr Cash/Bank (1000), Cr Accounts Receivable — AP1c GL
Customer Advance	Payment received before production or invoice	Dr Cash/Bank (1000), Cr Customer Advances (2500)
Staff Credit Repayment	Staff cash repayment of material purchase credit	Dr Cash/Bank (1000), Cr Staff Obligation Ledger

8.3 Bank Deposit Workflow

Unlinked bank credits appearing on the bank statement are recorded in the Bank Deposit pool (BD-XX-YY-NNNN), then matched to their source:

Status	Meaning
open	Deposit registered, not yet linked to any source
reserved	Temporarily reserved (30-minute window) to prevent double-linking
partial	Partially linked — some amount matched, remainder outstanding
allocated	Fully linked to customer receipts or advances
reclassified	Reclassified to non-sales category: expense_offset / inter_branch / refund_out / other_income
reversed	Reversed — bank error or returned transfer

8.4 Receipt Posting Rules

- Cannot post to a closed accounting period
- Cannot post a zero-amount receipt
- Duplicate detection: same quotation / amount / date combination is rejected
- Refund payer and approver must be different individuals (assertRefundPayerNotApprover)
- Cashier confirmation is mandatory before a receipt enters the ledger

9. FINANCE — ACCOUNTING DESK

9.1 GL Account Structure

Code	Account Name	Type
1000	Cash / Bank	Asset
1200	Staff Loans Receivable	Asset
1300	Inventory / Inter-Branch / Closing Stock	Asset
1500	Plant & Machinery (Capex)	Fixed Asset
1501	Land & Buildings (Capex)	Fixed Asset
1502	Furniture & Fittings (Capex)	Fixed Asset
1504	Generator (Capex)	Fixed Asset
2000	Accounts Payable — Suppliers	Liability
2400	Pension Payable	Liability
2500	Customer Advances / Refunds Payable	Liability
3200	Chairman Withdrawal (Equity Draw)	Equity
4000	Sales Revenue	Income
5010	Fuel & Lubricant (COGS)	COGS
5020	Maintenance (COGS)	COGS
5030	Outside Corrugation (COGS)	COGS
5040	Production Cost / Accessories (COGS)	COGS
5050	Purchases / Carriage Inward / Truck & Mining (COGS)	COGS
6000	Wages	Operating Expense
6100	Depreciation	Operating Expense
6110	Admin Salary	Operating Expense
6120	Admin / Office / Welfare / Security / HQ / Others	Operating Expense
6130	Rent & Utilities	Operating Expense
6140	IT & Software	Operating Expense
6150	Insurance	Operating Expense
6160	Professional Fees / Tax	Operating Expense
6170	Bank Charges	Operating Expense
6180	Zakat & Sallah	Operating Expense
6300	Interest Expense	Operating Expense
6400	Marketing & Advertising	Operating Expense

9.2 Expense Category → GL Mapping

Expense Category	GL Code	Special Rule
Wages	6000	
Fuel & lubricant	5010	
Maintenance	5020	
Outside corrugation	5030	
Production cost / Accessories	5040	
Purchases / Carriage inward / Truck & mining	5050	All map to same COGS account
Admin expenses / Office / HQ / Welfare / Security / Others / Miscellaneous	6120	Catch-all overhead
Admin salary	6110	
Rent & utilities	6130	

Expense Category	GL Code	Special Rule
IT & software	6140	
Insurance	6150	
Professional fees / Tax	6160	
Bank charges	6170	
Zakat & Sallah	6180	
Interest	6300	
Marketing & advertising	6400	
Pension	2400	Posts to PENSION PAYABLE (liability) — NOT an expense account
Staff loan	1200	Posts to STAFF LOANS RECEIVABLE (asset) — NOT an expense
Chairman withdrawal	3200	EQUITY DRAW — not an operating expense
Depreciation	6100	Non-cash; posted automatically by depreciation run
Land and buildings	1501	CAPEX — auto-registers fixed asset on full payment; useful life 240 months
Plant and machinery	1500	CAPEX — auto-registers fixed asset; useful life 60 months
Furniture & fittings	1502	CAPEX — auto-registers fixed asset; useful life 84 months
Generator	1504	CAPEX — auto-registers fixed asset; useful life 60 months

9.3 "Others" Expense Controls

Control	Rule
Minimum justification	Free-text explanation required (minimum 40 characters, configurable in org_policy_kv)
Finance review flag	Amounts above ₦50,000 flagged for Finance review (configurable)
AP3 alert	Amounts above ₦100,000 trigger AP3 unclassified cost alert (configurable)
Branch coaching alert	If Others exceeds 15% of total branch expenses, coaching alert triggered (configurable)

9.4 Accounting Phase Timeline

Phase	Name	Key Additions
AP1c	Production Recognition GL	Revenue recognised at production completion; GL auto-posted for receipts and production events
AP2	Received-Basis AP + Inventory Valuation	Payables recognised at GRN; inventory valued at landed cost; supplier advance tracking
AP3	Branch P&L + Material Costing + Payroll Labour	Full branch profit & loss; material cost classification; payroll labour allocated per production job

[IMPORTANT] Each phase is additive — AP3 requires AP2 which requires AP1c. Phase migration must be co-ordinated between finance_manager and MD.

9.5 Bank Reconciliation

Performed at /accounts/bank-reconciliation. Bank statement lines are imported, fingerprinted for duplicates (bankReconFingerprint.js), and matched against treasury movements. The control tie-out report validates that the GL cash balance matches the reconciled bank balance.

9.6 Debtors, Creditors & Write-Offs

The Accounting Desk provides debtors (AR) and creditors (AP) sub-ledger views. Receivable write-offs require MD approval above configured thresholds (receivableWriteOffPolicy.js). Settlements go through the accounting register settlement workflow: propose → review → pay, with each step tracked.

10. TREASURY & BANKING

10.1 Treasury Accounts & Reserve Policy

Each branch operates multiple treasury accounts (cash safes, bank accounts). The MD sets a Treasury Reserve Policy — a minimum balance for designated accounts — managed via /exec. Accounts below the reserve minimum are flagged on the Executive Dashboard.

10.2 Inter-Branch Loans

Fund transfers between branches are recorded as inter-branch loans to avoid distorting individual branch P&L. Both the lending and borrowing branches see the loan. Repayments are tracked until settlement. Requires Branch Manager or MD authorisation.

10.3 Supplier Advances (AP2)

Pre-payments to suppliers before goods receipt are tracked as supplier advances under AP2. On GRN, the advance is applied against the payable. AP2 supplier diagnostics identify unmatched advances outstanding.

10.4 Fixed Asset Depreciation

Depreciation is posted by an automated run (depreciationRunOps.js):

- Dr: Depreciation expense (GL 6100)
- Cr: Accumulated Depreciation (contra-asset linked to the fixed asset)

Disposal (fixedAssetDisposalOps.js) clears the asset from the register and posts the appropriate gain or loss entry.

11. EXECUTIVE DASHBOARD & COMMAND CENTRE

11.1 Access & Periods

The Executive Command Centre (/exec) is restricted to md, ceo, chairman, admin. Period options:

Period Key	Date Range
today	Current calendar day
week	Last 7 days including today
month (default)	Current calendar month — first to last day
last_month	Full previous calendar month
custom	User-specified start and end dates

11.2 Dashboard Panels

Panel	Data Displayed
Sales Summary	Revenue, collections, active/new quotations per period by branch
Expense Summary	Total by category; productive vs. non-productive split; branch breakdown
Receivables	Outstanding customer balances, overdue amounts, days outstanding
Branch Scorecard	Revenue, collections, expenses, gross margin per branch
BI Analytics	SKU weeks-cover (4-month lookback); cash horizon estimates
Working Capital	Cash position, payables, receivables, net working capital snapshot
MD Cockpit Pulses	Operational alerts: low stock, overdue deliveries, stale quotations, blocked coils
Champion Customers	Top customers by revenue and collections in the period
Material Costing (AP3)	Standard vs. actual material cost variance
Staff Activity	HR activity summary: pending approvals, attendance exceptions
Target vs. Actual	Branch sales targets vs. actual revenue
MD Attention Inbox	All pending decisions: credit exceptions, price exceptions, refunds, staff requests, work items
Stock Register Inbox	Stock register stages awaiting MD approval
Payroll Sign-Off Queue	Payroll runs awaiting mandatory MD approval
Bank Reconciliation Status	Unreconciled periods flagged company-wide
Reserve Policy	Treasury accounts below the MD-set minimum reserve balance

11.3 Decision Queue (/exec?tab=decide)

All pending decisions aggregated here, sorted by approval tier (critical → urgent → normal) and age. Includes: credit exceptions, below-floor price approvals, staff loan/leave requests, refund approvals, work item decisions, cutting list overrides, staff purchase credit requests, inter-branch requests.

12. OFFICE MANAGEMENT & WORK ITEMS

12.1 Office Threads & Memos

Office threads are structured memo/message chains between departments. Each thread can:

- Convert to a Payment Request (expense authorisation)
- Convert to a Material Request (stock transfer)
- Be filed in the Office Dossier (document cabinet)
- Link to Work Items for formal approval tracking
- Carry attachments with full version history
- Receive AI memo-polishing assistance (/api/office/ai/polish-memo)

12.2 Work Items & SLA

Priority	SLA Window
Critical	4 hours
Urgent / High	24 hours
Normal	48 hours

Work items are routed by role to the correct office desk:

Role	Office Desk
sales_manager	branch_manager
sales_staff	sales
operations_officer	operations
finance_manager / cashier	finance
md	executive
admin	office_admin

12.3 Official Notices

Managed by senior staff (md, admin, hr_admin, gmhr, ceo, chairman) or anyone with notices.manage. Notices can target all staff, specific branches, specific role keys, or departments. Options: require acknowledgement, pin to board, set expiry date, attach files.

12.4 Company Forum

- Branch scope — visible to the specific branch only
- Company scope — visible to all staff; only senior roles can create company-wide topics

12.5 Confidential Access

Work items and threads can be marked confidential. Only participants (userIsWorkItemParticipant) or staff with elevated access (workspaceConfidentialAccess.js) can view them. Used for sensitive HR and financial matters.

13. HUMAN RESOURCES — CORE

13.1 HR Module Areas

Area	URL	Primary Users
Self-Service	/me/* or /my-profile/*	All staff (including hr_portal_only)
Team HR	/team-hr/*	Branch Manager (sales_manager)
HR Administration	/hr/*	hr_admin, gmhr
Executive HR	/executive-hr/*	md, ceo, chairman

13.2 Staff Master File — Required Fields

Staff file completeness is assessed by `assessStaffFileCompleteness()`:

- Full name
- Date of birth
- Phone number
- Branch / site
- Job title / designation
- Date joined
- Bank account details
- Next of kin (name + phone)
- Highest academic qualification
- Probation end date (permanent staff only)

13.3 Designations & Functional Desks

Code	Title	Desk
desig_md	Managing Director	executive
desig_edo	Executive Director, Operations	executive
desig_edf	Executive Director, Finance	executive
desig_edc	Executive Director, Commercial	executive
desig_csec	Company Secretary	executive
desig_eamd	Executive Assistant to MD	executive
desig_gm	General Manager	executive
desig_gmhr	GM, Human Resources	hr
desig_bm	Branch Manager	branch
desig_actbm	Acting Branch Manager	branch
desig_hrrep	HR Representative	hr
desig_aud	Auditor	finance
desig_po	Procurement Officer	procurement
desig_hop	Head of Procurement	procurement
desig_pm	Production Manager	production
desig_qco	Quality Control Officer	production
desig_mm	Maintenance Manager	maintenance
desig_mtech	Maintenance Technician	maintenance

13.4 Leave Types

Type	Balance Tracked?	Entitlement / Rules
Annual	Yes	Junior (L1-L3): 14 days/year. Senior (L4-L7): 21 days/year. Accrued via leave accrual ledger.
Sick	No	No fixed balance; medical

Type	Balance Tracked?	Entitlement / Rules
		certificate required for extended periods
Maternity	Yes	Days configured via org_policy_kv (maternityLeaveDays)
Compassionate	No	Case-by-case — bereavement, family emergency
Leave without pay	No	Requires GMHR or MD approval (requiresGmHrApproval = true)
Other	No	Special circumstances not covered by other types

13.5 Transfer Tenure Policy (enforced in transferTenurePolicy.js)

Transfer Type	Minimum Service	Override Path
Branch transfer (inter-branch, HQ ↔ branch)	3 years	MD exception memo required
Internal rotation (non-branch)	2 years	GMHR or MD exception required
Branch Manager transfer	No minimum enforced separately	MD judgement

13.6 Salary Levels & Annual Leave

Level	Band	Annual Leave
L1	Junior	14 days
L2	Junior	14 days
L3	Junior	14 days
L4	Senior	21 days
L5	Senior	21 days
L6	Senior	21 days
L7	Senior	21 days (MD uses senior entitlement per board policy)

Default probation period: 6 months from date joined (PROBATION_MONTHS_DEFAULT). HR alerted when approaching end of probation.

14. HUMAN RESOURCES — PAYROLL & BENEFITS

14.1 Payroll Sign-Off Sequence

1. HR Admin prepares and submits payroll run for the period.
2. GMHR reviews and endorses the run.
3. MD approves — MANDATORY GATE (hr.payroll.md_approve). Hard-coded; cannot be bypassed.
4. Finance processes payment and exports bank transfer list (hr.payroll.pay + hr.payroll.export).

[CRITICAL] Payroll cannot be paid without the MD's explicit approval. This gate is enforced in code and cannot be waived by configuration.

14.2 Staff Obligation Ledger (OBL-XX-YY-NNNN)

A unified ledger tracks all staff financial obligations — loans, purchase credits, incident recoveries:

Transaction Type	Description
opening_balance	Initial balance when obligation account is created
disbursement	Loan or purchase credit issued to staff
payroll_deduction	Monthly deduction via payroll run (hr_payroll_line_loans table)
cash_repayment	Direct cash repayment at cashier desk
purchase_recognition	Material purchase credit recorded as obligation
adjustment	Manual correction or partial waiver (requires hr.recovery.manage)
write_off	Bad debt write-off (requires MD approval)

14.3 PAYE Tax & Pension

PAYE is computed per staff per period and exported for bank filing. Pension contributions (employer and employee portions) post to GL 2400 (Pension Payable liability, not an expense). Payroll export includes both PAYE and pension schedules.

14.4 Executive Benefits

- Director emoluments and retainers
- Vehicle and housing allowances
- Exceptional loans (exceptional_loan.approve — MD or Chairman required)
- Special increments (special_increment.approve)
- Director corporate eligibility: boardMember flag, or designation in (desig_md, desig_edo, desig_edf, desig_edc), or director_emolument compensation type

14.5 Chairman Family Accounts

A Chairman-specific module (hr.chairman.manage permission) tracks school fees payments for Chairman/CEO dependents, family expense payments, domestic staff management, and scholarship payments for staff/dependents. Accessible via /executive-hr/* to ceo and chairman roles.

15. DISCIPLINE, EXIT & ACCOUNTABILITY

15.1 4-Stage Discipline Ladder (Board Resolution)

Stage	Action	Owner	System Effect
1 — Verbal Warning	Conversation documented; note placed in HR file	Immediate Supervisor	Performance flag / negative record created; incident linked if applicable
2 — Written Warning / Query	Query letter issued; employee must respond in writing	Supervisor + HR	Discipline case created; letter workflow triggered; response tracked in case
3 — Final Warning or Suspension	Suspension with dates; payroll block may apply (MD-approved)	GMHR	Case escalated; suspension dates and payroll block status in system
4 — Demotion / Transfer / Separation	Formal separation, demotion, or transfer	MD + GMHR	Exit clearance workflow initiated; effective date and reason recorded

[CRITICAL] File tampering (removing/altering pages, backdating documents) is gross misconduct — triggers immediate query. Staff files retained for minimum 7 years.

15.2 Discipline Case Types

query, verbal_warning, written_warning, final_warning, suspension, investigation, gross_misconduct, negligence, absenteeism, lateness, harassment_complaint, insubordination, theft_fraud, property_damage, confidentiality_breach, policy_violation, performance_misconduct, dismissal_recommendation.

15.3 Discipline Case Status Flow

draft → open → awaiting_employee_response → under_investigation → awaiting_hr_review → awaiting_management_decision → action_issued → appealed (if contested) → closed / cancelled

15.4 Employee Exit — 5-Stage Clearance

Stage	Action	Permission Required
1 — Initiate	Exit initiated with reason and effective date	hr.exit.initiate (HR Admin)
2 — HR Clearance	ID returned, outstanding loans noted, documents collected	hr.exit.view (HR Admin)
3 — Finance Clearance	All obligations settled or waived; payroll finalised	hr.exit.finance_clear (financeHr bundle)
4 — Admin Clearance	Assets returned, access revoked, handover complete	hr.exit.admin_clear (HR Admin)
5 — Final Clearance	GMHR or MD gives final approval; exit record closed	hr.exit.final_clear (gmhr / MD)

[CRITICAL] All 5 stages must be completed before the employee's ERP account can be deactivated. The system blocks deactivation while any stage is pending.

15.5 Asset Custody Events

Event	Description
assign	Asset assigned to a staff member

Event	Description
transfer	Custody transferred from one staff to another
confirm_present	Physical verification that staff member still holds the asset
report_missing	Asset reported missing — triggers material incident and recovery schedule

15.6 Incident Recovery

1. Material incident linked to discipline case.
2. Involved parties recorded with responsibility weights.
3. Recovery schedule generated (hrIncidentRecoveryOps) for cost deduction from salary.
4. Recovery deductions linked to staff obligation account and processed via payroll.

16. ASSET & MAINTENANCE MANAGEMENT

16.1 Fixed Asset Register

Assets are created manually by finance_manager, or auto-created when a capex expense is fully paid (fixedAssetAutomationOps.js). Each asset carries: category, GL code, acquisition cost, acquisition date, useful life, accumulated depreciation, net book value, and disposal status.

16.2 Machine Tracking

- Machines tracked in machines table (MACH-YY-NNNN)
- Linked to fixed asset records (machine_asset_links)
- Meter readings tracked (machine_meter_logs) for usage monitoring
- Maintenance Manager (desig_mm) and Maintenance Technician (desig_mtech) manage the machine registry

16.3 Maintenance System

Entity	ID Format	Description
Maintenance Plan	MXPL-YY-NNNN	Recurring scheduled maintenance (e.g. weekly oil check, monthly belt inspection)
Maintenance Work Order	MXWO-YY-NNNN	Specific repair or maintenance job, from a plan or ad-hoc
Maintenance Event	MXEV-YY-NNNN	Completion record — date, technician, parts and labour costs

Maintenance cost lines classify under "Maintenance" expense (GL 5020) for production cost purposes.

16.4 Asset Disposal

1. Finance manager initiates disposal with sale/scrap value and disposal date.
2. GL posted: Dr Accumulated Depreciation + Dr Cash, Cr Asset Cost, Cr/Dr Gain or Loss.
3. Asset removed from active register; disposal reason recorded.

17. REPORTING & BUSINESS INTELLIGENCE

17.1 Standard Reports (/reports — requires reports.view permission)

Category	Key Reports
Sales	Sales by period/customer/product/gauge, quotation pipeline, cutting list status, delivery status
Finance	Cash flow, expense by category, trial balance, GL activity, debtor aging, creditor aging, bank reconciliation summary
Production	Output by machine/job, coil consumption, conversion variance, WIP status, stone-coated production report
Purchases	Purchase by supplier, PO status, GRN register, supplier advances, inventory valuation
Stock	Stock movement register, opening/closing stock, coil inventory, accessory stock, stone inventory, stock register sign-off status
HR	Staff list, attendance register, leave balances, loan outstanding, payroll summary, headcount by branch

17.2 Business Intelligence (/analytics)

- SKU weeks-cover (4-month lookback production demand)
- Revenue trends and gross margin analysis by branch and product
- Customer segmentation — champions, at-risk, dormant
- Cash horizon projections
- Branch comparative performance

17.3 Month-End Close Checklist

1. Stock Register — 4-stage sign-off completed and locked for all branches.
2. Bank Reconciliation — all treasury accounts reconciled to bank statement.
3. Payables Reconciliation — all supplier payables matched to GRNs and POs.
4. Receivables Review — debtor aging reviewed; write-offs approved by MD.
5. Depreciation Run — all fixed asset depreciation journals posted.
6. GL Trial Balance — all accounts in balance; no unexplained differences.
7. Period Locked — accounting period closed; no further postings accepted.

17.4 Accounting Phase Reports

Phase	Key Reports
AP1c	Production recognition GL log, receipt GL activity, customer advance reconciliation, refund GL activity
AP2	Supplier advance report, inventory GL alignment, received-basis AP rebuild preview, supplier diagnostics
AP3	Material cost report (standard vs actual), costing readiness check, branch P&L, cost classification breakdown

18. FILING, AUDIT TRAIL & DOCUMENT STANDARDS

18.1 Numbering Principles

All sequences stored in `human_id_sequences` table, incremented atomically. Sequences reset each calendar year per branch. The `branches` table stores the short branch code (KD, YL, MDG) used in IDs.

18.2 Print Snapshots

Point-in-time snapshots are captured for: stock register (at lock stage), payroll (at MD approval), work items (at closure). Stored in `work_item_print_snapshots`. These protect against retrospective document alteration.

18.3 Audit Trail

Every significant action is logged in `audit_log` with: actor ID and name, timestamp, action type, target entity and ID, branch ID. HR actions are additionally in `hr_audit_events`. Custom permission changes in `customPermissionAudit`. All audit records are immutable — they cannot be deleted or modified.

18.4 Filing Completeness

Filing completeness (`filingCompleteness.js`) is checked before a work item can be formally closed. Documents without a filing number cannot be archived. AI-assisted auto-filing is available (`/api/office/threads/:id/ai-file`).

19. AI ASSISTANT — ZARE

19.1 Overview

Zarewa ERP includes a built-in AI assistant named Zare. Zare assists staff with ERP navigation, workflow explanations, memo drafting, and operational Q&A. Accessible via the in-app Help/Chat interface.

19.2 Capabilities

- Natural language Q&A on ERP workflows (intent detection with typo tolerance)
- Memo drafting assistance (helpMemoAssist.js, smartMemoComposer.js)
- Office memo polishing (/api/office/ai/polish-memo)
- Transaction-specific contextual help and error message explanation
- Workflow coaching for blocked actions
- ERP query interpretation — natural language to structured data query (helpErpQuery.js)
- Business analysis insights for MD and senior staff (helpBusinessAnalysis.js)

19.3 Personalised Memory

- User memory (help_user_memory) — remembers individual preferences and frequent queries
- Branch memory (help_branch_memory) — captures branch-level operational context

Memory persists across sessions, enabling increasingly contextual and relevant responses.

19.4 RAG Knowledge Base

Zare uses Retrieval-Augmented Generation with a knowledge base in help_rag_chunks (weighted by help_article_weights). Knowledge gaps — queries Zare could not answer — are logged in help_knowledge_gaps for human review and knowledge base improvement (helpSelfTrain.js).

20. SYSTEM ADMINISTRATION

20.1 Admin Responsibilities

- User account creation, role assignment, and password reset
- Custom permission grants and revocations per user (on top of role bundles)
- Branch configuration and pricing policy management
- Org governance limits configuration (credit thresholds, expense review thresholds)
- System monitoring at /workspace/monitoring
- Database migration execution (migrate.js)

20.2 Org Policy Settings (stored in org_policy_kv with audit trail)

Policy Key	Description	Default
CREDIT_BRANCH_MANAGER_LIMIT_NGN	Max credit amount BM can approve alone	Not set
CREDIT_MD_REQUIRED_ABOVE_NGN	Threshold above which MD approval is mandatory	Not set
DELIVERY_PAYMENT_GATE	Gate mode: off / warn / enforce	off
ALLOW_MD_DELIVERY_OVERRIDE	MD can bypass enforce mode per delivery	false
DELIVERY_PAYMENT_GATE_STRICT_FINANCE	Finance must confirm before dispatch	false
SESSION_TIMEOUT_MINUTES	Session idle timeout (range 5-480)	120
FAILED_LOGIN_LOCK_THRESHOLD	Failed logins before account lock	5
ACCOUNT_LOCK_MINUTES	Account lock duration in minutes	30
OTHERS_MIN_JUSTIFICATION_LEN	Min chars for Others expense justification	40
OTHERS_FINANCE_REVIEW_THRESHOLD_NGN	Others amount flagging Finance review	50000
AP3_UNCLASSIFIED_ALERT_THRESHOLD_NGN	Unclassified amount triggering AP3 alert	100000
OTHERS_BRANCH_COACH_THRESHOLD_OLD_PCT	Others % of branch expenses for coaching alert	15
maternityLeaveDays	Maternity leave entitlement in days	Per policy

20.3 Regular Maintenance

- Daily: Confirm server uptime; review audit log; check failed login attempts
- Weekly: Verify database backup integrity; review user session activity
- Monthly: Database integrity check; review custom permission changes; verify stock register progress across branches
- Quarterly: Rotate admin credentials; audit user access list; test disaster recovery procedures

APPENDIX A — COMPLETE EXPENSE CATEGORY & GL REFERENCE

Category	GL	Type	Notes
Wages	6000	COGS	Direct production wages
Fuel & lubricant	5010	COGS	Production/operations fuel
Maintenance	5020	COGS	Plant and equipment maintenance
Outside corrugation	5030	COGS	Sub-contracted corrugation services
Production cost	5040	COGS	Direct production costs
Purchases	5050	COGS	Raw coil materials
Accessories	5040	COGS	Consumables and accessories for production
Carriage inward	5050	COGS	Inbound freight and haulage
Closing stock	1300	Asset	Monthly inventory balance entry
Truck & mining	5050	COGS	Vehicle, transport, mining operations
Admin expenses	6120	OpEx	General administration costs
Admin salary	6110	OpEx	Non-production staff salaries
Bank charges	6170	OpEx	Bank fees and transfer charges
Rent & utilities	6130	OpEx	Premises, power, water
Office expenses	6120	OpEx	Stationery, office supplies
IT & software	6140	OpEx	Technology subscriptions, software
Insurance	6150	OpEx	Insurance premiums
Professional fees	6160	OpEx	Legal, audit, consulting fees
Tax	6160	OpEx	Business taxes and levies
Pension	2400	LIABILITY	Posts to Pension Payable — NOT expense
Security	6120	OpEx	Guards and security systems
Interest	6300	OpEx	Loan and finance charges
Welfare	6120	OpEx	Staff welfare and amenities
Zakat & Sallah	6180	OpEx	Religious obligations and festivals
Marketing & advertising	6400	OpEx	Marketing and promotional spend
HQ	6120	OpEx	Head office overhead allocation
Chairman withdrawal	3200	EQUITY	Equity draw — NOT an operating expense
Staff loan	1200	ASSET	Posts to Staff Loans Receivable — NOT expense
Others	6120	OpEx	Min 40-char justification required; Finance review if >N50k

Category	GL	Type	Notes
Miscellaneous	6120	OpEx	Use sparingly; same controls as Others
Depreciation	6100	OpEx	Non-cash; posted automatically by depreciation run
Land and buildings	1501	CAPEX	Auto-registers fixed asset on full payment; 240-month useful life
Plant and machinery	1500	CAPEX	Auto-registers fixed asset; 60-month useful life
Furniture & fittings	1502	CAPEX	Auto-registers fixed asset; 84-month useful life
Generator	1504	CAPEX	Auto-registers fixed asset; 60-month useful life

APPENDIX B — HR ROLE PERMISSION BUNDLES

Bundle	Roles	Key Permissions Included
selfService	All staff (default)	hr.self, my_profile.view, my_leave.request, my_loan.request, my_attendance.view, my_payslip.view, my_documents.view
branchManager	sales_manager	selfService + hr.team.view, attendance.mark, daily_roll.mark, leave.endorse, loan.endorse, branch.endorse_staff, incident.create, transfer.recommend, absence.view/manage
hrAdmin	hr_admin	Full HR: staff.manage, all 5 exit clearance stages, asset custody, recovery management, all leave/loan/discipline management
gmhr	gmhr	gm_approve, final_approve, exit.final_clear, letters.approve, transfer final approval
mdExecutive	md	executive.view, payroll.md_approve, salary_structure.approve, exceptional_loan.approve, chairman.manage, special_increment.approve, special_beneficiary.manage, branch_contribution.mark
executiveScholarshipDomestic	ceo, chairman	executive.view + benefits, chairman.manage, special_beneficiary.manage, payroll.view_sensitive, payroll.md_approve, directory.view, staff.manage, hq.view_all_branches
financeHr	finance_manager (sub-bundle)	hr.payroll.pay, hr.payroll.export, hr.loan.disburse, hr.exit.finance_clear, hr.exit.view

APPENDIX C — GLOSSARY

Term	Definition
AP1c	Accounting Phase 1c — production-completion revenue recognition with automatic GL posting
AP2	Accounting Phase 2 — received-basis accounts payable; inventory valued at landed cost
AP3	Accounting Phase 3 — branch P&L, material cost classification, payroll labour per production job
COGS	Cost of Goods Sold — direct production costs (GL accounts 5010–5050)
Credit Exception	Formal approval allowing a customer to receive goods on credit with agreed terms
Cutting List	Production work order (CL-XX-YY-NNNN) issued from Sales to the factory
Delivery Gate	Payment enforcement check before goods are dispatched (modes: off / warn / enforce)
Finance Desk Separation	Phase B control — Cashier desk (/cashier) and Accounting desk (/accounting) separated by permission
GRN	Goods Received Note — records coil or accessory arrival at a branch from supplier
Obligation Account	Staff financial obligation ledger (OBL) tracking loans, purchase credits, and recoveries
Period Control	System rule blocking financial postings to closed accounting periods
Policy V1	Revenue recognised at production completion — not at delivery or invoice date
RAG	Retrieval-Augmented Generation — AI knowledge base architecture powering Zare
RBAC	Role-Based Access Control — Zarewa's permission system with 100+ fine-grained keys
Reserve Policy	MD-set minimum cash balance for designated treasury accounts
Staff Obligation	Unified ledger for all staff financial obligations (loans, credits, recoveries, write-offs)
WIP	Work-in-Progress — goods currently in production; not yet completed or delivered
Zare	Zarewa's built-in AI assistant for ERP navigation, memo drafting, and operational guidance